

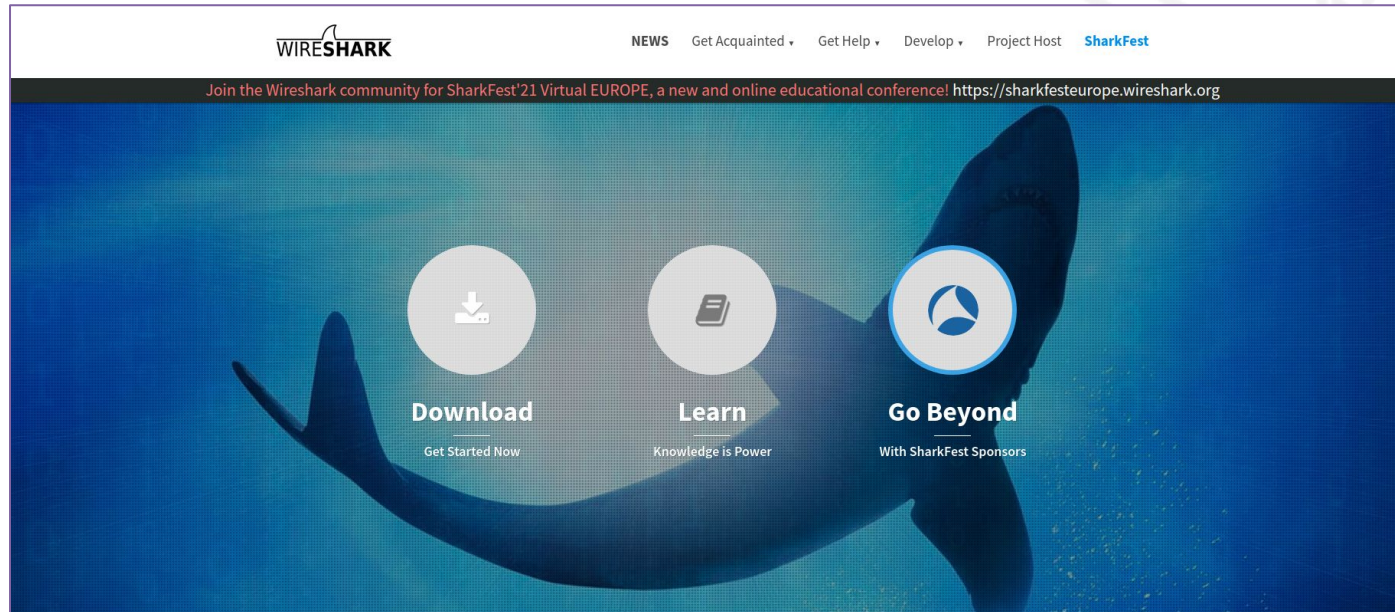
Introdução ao Wireshark

Redes de Computadores

Introdução ao Wireshark

- Programa gráfico que analisa e filtra o tráfego de uma rede, organizando-o por protocolos
- Pode monitorar a entrada/saída de dados de um computador
- Sniffer: programa usado para detectar problemas, conexões suspeitas, testar criptografias e outras atividades da rede
- Endereço web: www.wireshark.org

Download do Wireshark



www.wireshark.org

Exercício Resolvido (1): Identificando seu IP

- Abra o terminal da sua máquina e identifique seu número de IP. No Windows, use o comando `ipconfig` e, no Linux, `ifconfig`

Exercício Resolvido (1): Identificando seu IP

- Abra o terminal da sua máquina e identifique seu número de IP. No Windows, use o comando ipconfig e, no Linux, ifconfig

```
~$ ifconfig
enp1s0: flags=4099<UP,BROADCAST,MULTICAST> mtu 1500
    ether 74:e6:e2:d0:48:f7 txqueuelen 1000 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

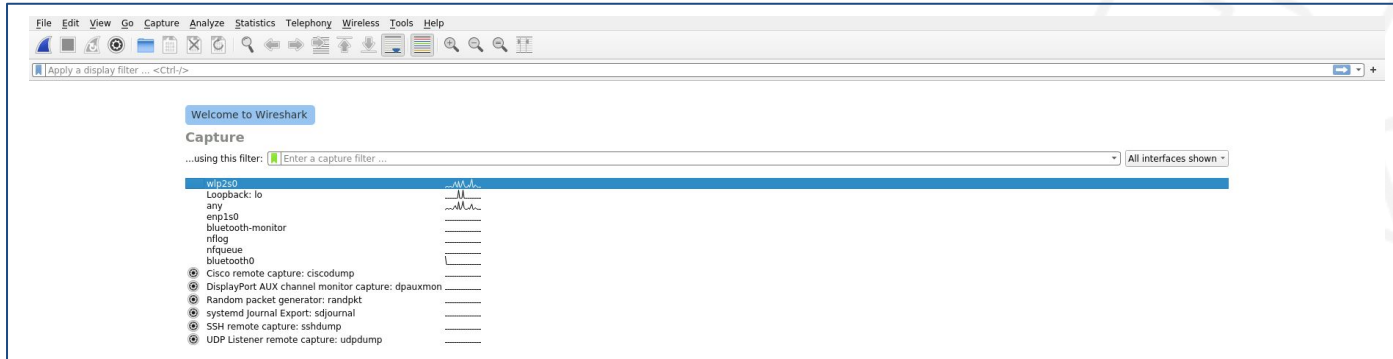
lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Loopback Local)
    RX packets 7139 bytes 836170 (836.1 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 7139 bytes 836170 (836.1 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

wlp2s0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.0.5 netmask 255.255.255.0 broadcast 192.168.0.255
    inet6 2804:14c:5b75:9b5d:bde:70cb:ca63:78b6 prefixlen 64 scopeid 0x0<global>
    inet6 2804:14c:5b75:9b5d:7447:b9b9:a20b:f27 prefixlen 64 scopeid 0x0<global>
    inet6 fe80::5079:8cb:ba74:a045 prefixlen 64 scopeid 0x20<link>
    ether 38:b1:db:cc:f7:eb txqueuelen 1000 (Ethernet)
    RX packets 179280 bytes 173873568 (173.8 MB)
    RX errors 0 dropped 1909 overruns 0 frame 0
    TX packets 165606 bytes 120735335 (120.7 MB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

Resp: FIG

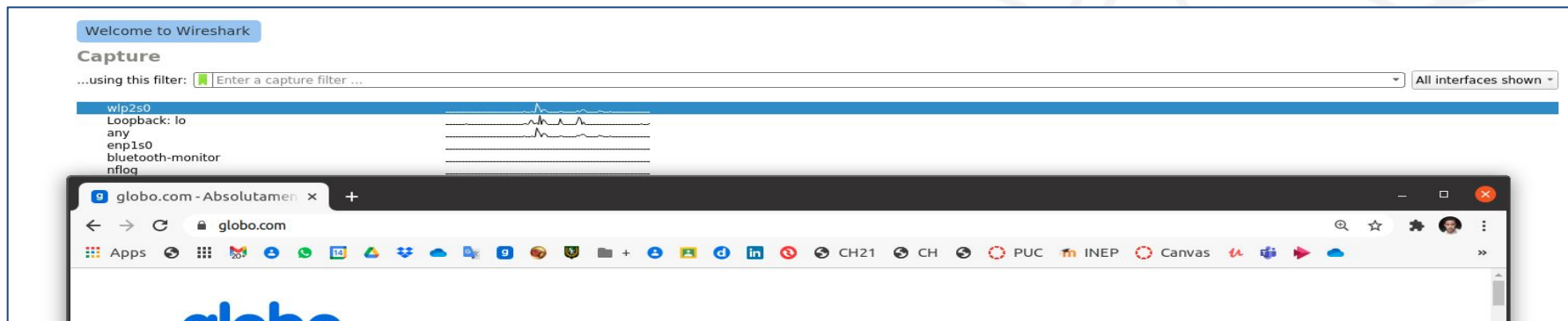
Exercício Resolvido (2): Tela Inicial

Resp: FIG

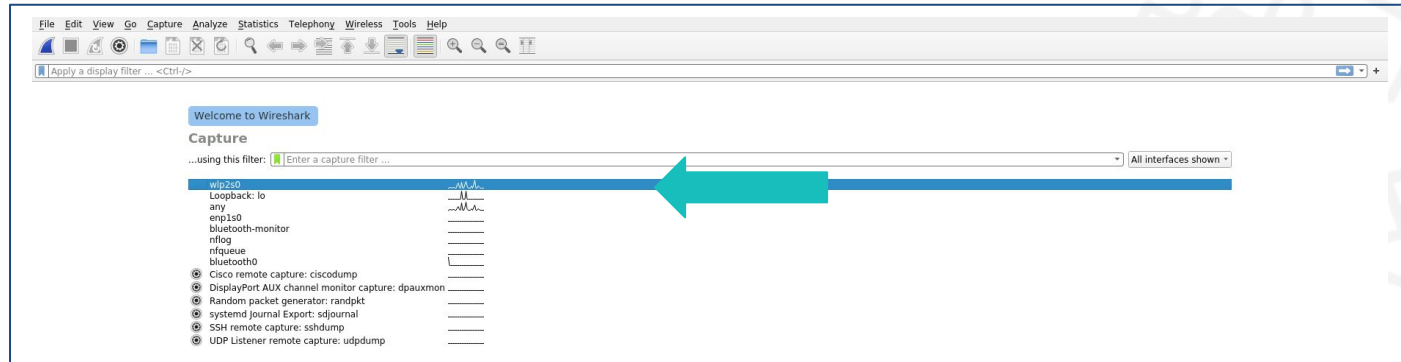


Exercício Resolvido (3): Abra um Navegador

Resp: FIG



Selecione uma Interface de Rede



Exercício Resolvido (4): Selecione uma Interface

Apply a display filter ... <Ctrl-/>						
No.	Time	Source	Destination	Protocol	Length	Info
575	95.459022295	2804:14c:5b75:9b5d:...	2800:3f0:4004:803:...	UDP	95	32796 → 443 Len=33
576	96.255957750	192.168.0.5	151.101.178.133	TCP	66	[TCP Dup ACK 77#2] 35146 → 443 [ACK] Seq=1 Ack=1 Win=501 Len=...
577	96.283229755	151.101.178.133	192.168.0.5	TCP	66	[TCP Dup ACK 79#2] [TCP ACKed unseen segment] 443 → 35146 [AC...
578	96.426704533	192.168.0.5	8.43.72.113	TCP	66	[TCP Previous segment not captured] 56852 → 443 [FIN, ACK] Se...
579	96.426808795	192.168.0.5	8.43.72.113	TCP	66	[TCP Previous segment not captured] 56846 → 443 [FIN, ACK] Se...
580	96.426859112	192.168.0.5	8.43.72.113	TCP	66	[TCP Previous segment not captured] 56850 → 443 [FIN, ACK] Se...
581	96.427428798	2804:14c:5b75:9b5d:...	2800:3f0:4004:807:...	UDP	984	48161 → 443 Len=922
582	96.427989143	192.168.0.5	181.213.132.2	DNS	88	Standard query 0x1f2d A beacons3.gvt2.com OPT
583	96.428368498	192.168.0.5	181.213.132.2	DNS	88	Standard query 0x3a9d AAAA beacons3.gvt2.com OPT
584	96.485042639	2800:3f0:4004:807:...	2804:14c:5b75:9b5d:...	UDP	87	443 → 48161 Len=25
585	96.568956990	8.43.72.113	192.168.0.5	TCP	66	[TCP ACKed unseen segment] 443 → 56846 [ACK] Seq=1 Ack=3 Win=...
586	96.569035633	8.43.72.113	192.168.0.5	TCP	66	443 → 56846 [FIN, ACK] Seq=1 Ack=3 Win=13756 Len=0 TSval=4734...
587	96.569072573	192.168.0.5	8.43.72.113	TCP	66	56846 → 443 [ACK] Seq=3 Ack=2 Win=501 Len=0 TSval=1608785144 ...
588	96.569107920	8.43.72.113	192.168.0.5	TCP	66	[TCP ACKed unseen segment] 443 → 56850 [ACK] Seq=1 Ack=3 Win=...
589	96.569130187	8.43.72.113	192.168.0.5	TCP	66	443 → 56850 [FIN, ACK] Seq=1 Ack=3 Win=14480 Len=0 TSval=4734...
590	96.569141734	192.168.0.5	8.43.72.113	TCP	66	56850 → 443 [ACK] Seq=3 Ack=2 Win=501 Len=0 TSval=1608785144 ...
591	96.569157826	2800:3f0:4004:807:...	2804:14c:5b75:9b5d:...	UDP	126	443 → 48161 Len=64
592	96.569200300	8.43.72.113	192.168.0.5	TCP	66	[TCP ACKed unseen segment] 443 → 56852 [ACK] Seq=1 Ack=3 Win=...
593	96.570217294	2804:14c:5b75:9b5d:...	2800:3f0:4004:807:...	UDP	111	48161 → 443 Len=49
594	96.574897079	8.43.72.113	192.168.0.5	TCP	66	443 → 56852 [FIN, ACK] Seq=1 Ack=3 Win=13756 Len=0 TSval=4734...
595	96.574951239	192.168.0.5	8.43.72.113	TCP	66	56852 → 443 [ACK] Seq=3 Ack=2 Win=501 Len=0 TSval=1608785150 ...
596	96.574987594	2800:3f0:4004:807:...	2804:14c:5b75:9b5d:...	UDP	87	443 → 48161 Len=25
597	96.600729374	2804:14c:5b75:9b5d:...	2800:3f0:4004:807:...	UDP	95	48161 → 443 Len=33
598	96.631616014	2800:3f0:4004:807:...	2804:14c:5b75:9b5d:...	UDP	87	443 → 48161 Len=25
599	97.486778658	8c:fd:de:87:bf:42	Spanning-tree-(for-...	STP	60	Conf. Root = 32768/0/8c:fd:de:87:bf:41 Cost = 0 Port = 0x80...
600	99.470731000	8c:fd:de:87:bf:42	Spanning-tree-(for-...	STP	60	Conf. Root = 32768/0/8c:fd:de:87:bf:41 Cost = 0 Port = 0x80...
601	101.375947604	HonHaiPr.cc:f7:eb	8c:fd:de:87:bf:41	ARP	42	Who has 192.168.0.1? Tell 192.168.0.5
602	101.382850271	8c:fd:de:87:bf:41	HonHaiPr.cc:f7:eb	ARP	60	192.168.0.1 is at 8c:fd:de:87:bf:41
603	101.430277693	192.168.0.5	181.213.132.3	DNS	77	Standard query 0x1f2d A beacons3.gvt2.com
604	101.430574597	2804:14c:5b75:9b5d:...	2804:14d:1:0:181:21...	DNS	97	Standard query 0x3a9d AAAA beacons3.gvt2.com
605	101.454666766	8c:fd:de:87:bf:42	Spanning-tree-(for-...	STP	60	Conf. Root = 32768/0/8c:fd:de:87:bf:41 Cost = 0 Port = 0x80...
606	101.773335878	8c:fd:de:87:bf:41	HonHaiPr.cc:f7:eb	ARP	60	Who has 192.168.0.5? Tell 192.168.0.1
607	101.773360971	HonHaiPr.cc:f7:eb	8c:fd:de:87:bf:41	ARP	42	192.168.0.5 is at 38:b1:db:cc:f7:eb
608	102.399971853	2804:14c:5b75:9b5d:...	2606:4700:6810:b73	TCP	74	[TCP Dup ACK 103#2] 57592 → 443 [ACK] Seq=1 Ack=1 Win=501 Len=...
609	102.496268386	2804:14c:5b75:9b5d:...	2800:3f0:4004:802:...	UDP	96	57407 → 443 Len=34
610	102.543897158	2800:3f0:4004:802:...	2804:14c:5b75:9b5d:...	UDP	88	443 → 57407 Len=26
611	102.571606461	2606:4700:6810:b73	2804:14c:5b75:9b5d:...	TCP	74	[TCP Dup ACK 106#2] [TCP ACKed unseen segment] 443 → 57592 [A...

Resp: FIG

Exercício Resolvido (5): Tráfego Destinado ao seu IP

- Na opção de filtro, selecione **ip.dst == seu-ip**

ip.dst == 192.168.0.5					
No.	Time	Source	Destination	Protocol	Length Info
21065	1183.2915512..	131.0.25.125	192.168.0.5	TLSv1.2	1514 Server Hello
21067	1183.3098107..	131.0.25.125	192.168.0.5	TLSv1.2	1514 Certificate [TCP segment of a reassembled PDU]
21069	1183.3135359..	131.0.25.125	192.168.0.5	TLSv1.2	299 Server Key Exchange, Server Hello Done
21074	1183.3500779..	131.0.25.125	192.168.0.5	TLSv1.2	340 New Session Ticket, Change Cipher Spec, Encrypted Handshake M...
21076	1183.3531966..	131.0.25.125	192.168.0.5	TCP	66 443 → 44024 [ACK] Seq=3404 Ack=3109 Win=24448 Len=0 TSval=233...
21077	1183.3818931..	131.0.25.125	192.168.0.5	TLSv1.2	901 Application Data
21086	1184.8594591..	151.101.177.44	192.168.0.5	TLSv1.3	117 Application Data
21089	1184.8633804..	151.101.177.44	192.168.0.5	TCP	66 443 → 46808 [FIN, ACK] Seq=21522 Ack=8279 Win=90624 Len=0 TS...
21091	1184.8943067..	151.101.177.44	192.168.0.5	TCP	66 443 → 46808 [ACK] Seq=21523 Ack=8280 Win=90624 Len=0 TSval=33...
21216	1194.9735189..	131.0.25.125	192.168.0.5	TCP	60 443 → 44024 [RST, ACK] Seq=4239 Ack=3109 Win=0 Len=0
21299	1260.9047769..	192.168.0.1	192.168.0.5	DHCP	342 DHCP ACK - Transaction ID 0x2c2c3357
21309	1260.9360920..	104.244.42.136	192.168.0.5	TCP	60 443 → 44024 [ACK] Seq=4445 Ack=1342 Win=...
21388	1262.1550528..	181.213.132.2	192.168.0.5	DNS	118 Standard query response 0xe003 A signaler-pa.clients6.google...
21389	1262.1550903..	181.213.132.2	192.168.0.5	DNS	130 Standard query response 0x3976 AAAA signaler-pa.clients6.goog...
21218	1242.1457633..	181.213.132.2	192.168.0.5	DNS	148 Standard query response 0x10b5 A connectivity-check.ubuntu.co...
22136	1242.3077891..	35.224.170.84	192.168.0.5	TCP	74 80 → 54592 [SYN, ACK] Seq=0 Ack=1 Win=64768 Len=0 MSS=1420 SA...
22140	1242.4718300..	35.224.170.84	192.168.0.5	TCP	66 80 → 54592 [ACK] Seq=1 Ack=88 Win=64256 Len=0 TSval=269574130...
22141	1242.4719109..	35.224.170.84	192.168.0.5	HTTP	214 HTTP/1.1 204 No Content
22143	1242.4720050..	35.224.170.84	192.168.0.5	TCP	66 80 → 54592 [FIN, ACK] Seq=149 Ack=88 Win=64256 Len=0 TSval=26...
22152	1242.6339716..	35.224.170.84	192.168.0.5	TCP	66 80 → 54592 [ACK] Seq=150 Ack=89 Win=64256 Len=0 TSval=2695741...
22207	1243.1643929..	131.0.25.125	192.168.0.5	TCP	74 443 → 44028 [SYN, ACK] Seq=0 Ack=1 Win=17896 Len=0 MSS=1460 S...
22210	1243.1971349..	131.0.25.125	192.168.0.5	TCP	66 443 → 44028 [ACK] Seq=1 Ack=525 Win=19072 Len=0 TSval=2333383...
22211	1243.1971837..	131.0.25.125	192.168.0.5	TLSv1.2	222 Server Hello, Change Cipher Spec, Encrypted Handshake Message
22216	1243.2282898..	131.0.25.125	192.168.0.5	TCP	66 443 → 44028 [ACK] Seq=157 Ack=1937 Win=21760 Len=0 TSval=2333...
22217	1243.2304755..	131.0.25.125	192.168.0.5	TLSv1.2	876 Application Data
22320	1247.2888994..	181.213.132.2	192.168.0.5	DNS	171 Standard query response 0x116c No such name A beacons.gcp.gvt...
22324	1247.3758445..	104.244.42.136	192.168.0.5	TCP	66 [TCP Keep-Alive ACK] 443 → 58224 [ACK] Seq=4445 Ack=1342 Win=...
22438	1253.5641292..	131.0.25.125	192.168.0.5	TCP	60 443 → 44028 [RST, ACK] Seq=967 Ack=3034 Win=0 Len=0
22864	1292.5249495..	104.244.42.136	192.168.0.5	TCP	66 [TCP Keep-Alive ACK] 443 → 58224 [ACK] Seq=4445 Ack=1342 Win=...
22918	1303.1598502..	181.213.132.2	192.168.0.5	DNS	171 Standard query response 0x5587 A horizon-track.globo.com CNAME...
22919	1303.1598895..	181.213.132.2	192.168.0.5	DNS	208 Standard query response 0x0afc AAAA horizon-track.globo.com C...
22921	1303.1826834..	181.213.132.2	192.168.0.5	DNS	149 Standard query response 0x984a AAAA horizon-track-b.globo.com C...
22923	1303.2104284..	131.0.25.125	192.168.0.5	TCP	74 443 → 44030 [SYN, ACK] Seq=0 Ack=1 Win=17896 Len=0 MSS=1460 S...
22926	1303.2372546..	131.0.25.125	192.168.0.5	TCP	66 443 → 44030 [ACK] Seq=1 Ack=525 Win=19072 Len=0 TSval=2571884...
22927	1303.2373110..	131.0.25.125	192.168.0.5	TLSv1.2	222 Server Hello, Change Cipher Spec, Encrypted Handshake Message
22932	1303.2774425..	131.0.25.125	192.168.0.5	TCP	66 443 → 44030 [ACK] Seq=157 Ack=1937 Win=21760 Len=0 TSval=2571...
22933	1303.2775025..	131.0.25.125	192.168.0.5	TLSv1.2	902 Application Data

Resp: FIG

Exercício Resolvido (6): Ping em 8.8.8.8

- Abra o terminal e execute o comando **ping 8.8.8.8**
- Na opção de filtro, escolha: **ip.addr == seu-ip and icmp.type == 8 and ip.dst == 8.8.8.8**

Exercício Resolvido (6): Ping em 8.8.8.8

ip.addr == 192.168.0.5 and icmp.type == 8 and ip.dst == 8.8.8.8						
No.	Time	Source	Destination	Protocol	Length	Info
43185	3397.3337096...	192.168.0.5	8.8.8.8	ICMP	98	Echo (ping) request id=0x0007, seq=1/256, ttl=64 (reply in 4..
43188	3398.3355593...	192.168.0.5	8.8.8.8	ICMP	98	Echo (ping) request id=0x0007, seq=2/512, ttl=64 (reply in 4..
43190	3399.3375130...	192.168.0.5	8.8.8.8	ICMP	98	Echo (ping) request id=0x0007, seq=3/768, ttl=64 (reply in 4..
43193	3400.3388582...	192.168.0.5	8.8.8.8	ICMP	98	Echo (ping) request id=0x0007, seq=4/1024, ttl=64 (reply in ..
43195	3401.3399109...	192.168.0.5	8.8.8.8	ICMP	98	Echo (ping) request id=0x0007, seq=5/1280, ttl=64 (reply in ..
43198	3402.3412689...	192.168.0.5	8.8.8.8	ICMP	98	Echo (ping) request id=0x0007, seq=6/1536, ttl=64 (reply in ..
43228	3403.3428465...	192.168.0.5	8.8.8.8	ICMP	98	Echo (ping) request id=0x0007, seq=7/1792, ttl=64 (reply in ..
43234	3404.3439657...	192.168.0.5	8.8.8.8	ICMP	98	Echo (ping) request id=0x0007, seq=8/2048, ttl=64 (reply in ..
43236	3405.3457662...	192.168.0.5	8.8.8.8	ICMP	98	Echo (ping) request id=0x0007, seq=9/2304, ttl=64 (reply in ..
43294	3406.3476088...	192.168.0.5	8.8.8.8	ICMP	98	Echo (ping) request id=0x0007, seq=10/2560, ttl=64 (reply in..
43302	3407.3479655...	192.168.0.5	8.8.8.8	ICMP	98	Echo (ping) request id=0x0007, seq=11/2816, ttl=64 (reply in..
43305	3408.3496919...	192.168.0.5	8.8.8.8	ICMP	98	Echo (ping) request id=0x0007, seq=12/3072, ttl=64 (reply in..
43307	3409.3513843...	192.168.0.5	8.8.8.8	ICMP	98	Echo (ping) request id=0x0007, seq=13/3328, ttl=64 (reply in..
43318	3410.3533855...	192.168.0.5	8.8.8.8	ICMP	98	Echo (ping) request id=0x0007, seq=14/3584, ttl=64 (reply in..
43322	3411.3545962...	192.168.0.5	8.8.8.8	ICMP	98	Echo (ping) request id=0x0007, seq=15/3840, ttl=64 (reply in..
43328	3412.3557194...	192.168.0.5	8.8.8.8	ICMP	98	Echo (ping) request id=0x0007, seq=16/4096, ttl=64 (reply in..
43331	3413.3559512...	192.168.0.5	8.8.8.8	ICMP	98	Echo (ping) request id=0x0007, seq=17/4352, ttl=64 (reply in..
43334	3414.3578621...	192.168.0.5	8.8.8.8	ICMP	98	Echo (ping) request id=0x0007, seq=18/4608, ttl=64 (reply in..
43336	3415.3593249...	192.168.0.5	8.8.8.8	ICMP	98	Echo (ping) request id=0x0007, seq=19/4864, ttl=64 (reply in..
43342	3416.3599450...	192.168.0.5	8.8.8.8	ICMP	98	Echo (ping) request id=0x0007, seq=20/5120, ttl=64 (reply in..
43344	3417.3610168...	192.168.0.5	8.8.8.8	ICMP	98	Echo (ping) request id=0x0007, seq=21/5376, ttl=64 (reply in..
43347	3418.3626304...	192.168.0.5	8.8.8.8	ICMP	98	Echo (ping) request id=0x0007, seq=22/5632, ttl=64 (reply in..
43349	3419.3640341...	192.168.0.5	8.8.8.8	ICMP	98	Echo (ping) request id=0x0007, seq=23/5888, ttl=64 (reply in..
43353	3420.3660993...	192.168.0.5	8.8.8.8	ICMP	98	Echo (ping) request id=0x0007, seq=24/6144, ttl=64 (reply in..
43355	3421.3675393...	192.168.0.5	8.8.8.8	ICMP	98	Echo (ping) request id=0x0007, seq=25/6400, ttl=64 (reply in..
43358	3422.3679463...	192.168.0.5	8.8.8.8	ICMP	98	Echo (ping) request id=0x0007, seq=26/6656, ttl=64 (reply in..
43366	3423.3697223...	192.168.0.5	8.8.8.8	ICMP	98	Echo (ping) request id=0x0007, seq=27/6912, ttl=64 (reply in..
43369	3424.3714386...	192.168.0.5	8.8.8.8	ICMP	98	Echo (ping) request id=0x0007, seq=28/7168, ttl=64 (reply in..
43371	3425.3719397...	192.168.0.5	8.8.8.8	ICMP	98	Echo (ping) request id=0x0007, seq=29/7424, ttl=64 (reply in..
43374	3426.3738848...	192.168.0.5	8.8.8.8	ICMP	98	Echo (ping) request id=0x0007, seq=30/7680, ttl=64 (reply in..
43376	3427.3756102...	192.168.0.5	8.8.8.8	ICMP	98	Echo (ping) request id=0x0007, seq=31/7936, ttl=64 (reply in..
43381	3428.3770527...	192.168.0.5	8.8.8.8	ICMP	98	Echo (ping) request id=0x0007, seq=32/8192, ttl=64 (reply in..
43385	3429.3788675...	192.168.0.5	8.8.8.8	ICMP	98	Echo (ping) request id=0x0007, seq=33/8448, ttl=64 (reply in..
43388	3430.3799811...	192.168.0.5	8.8.8.8	ICMP	98	Echo (ping) request id=0x0007, seq=34/8704, ttl=64 (reply in..
43392	3431.3819564...	192.168.0.5	8.8.8.8	ICMP	98	Echo (ping) request id=0x0007, seq=35/8960, ttl=64 (reply in..
43395	3432.3833493...	192.168.0.5	8.8.8.8	ICMP	98	Echo (ping) request id=0x0007, seq=36/9216, ttl=64 (reply in..

Resp: FIG

Exercício (1): Ping em 8.8.8.8

- No exercício anterior, podemos simplificar a expressão **ip.addr == 192.168.0.5 and icmp.type == 8 and ip.dst == 8.8.8.8**? Discuta sua resposta

Exercício (2): ICMP

- O que acontece se usarmos o filtro “**icmp**”?

Exercício Resolvido (7): ARP

- No filtro, selecione opção ARP

Exercício Resolvido (7): ARP

- No filtro, selecione opção ARP

arp						
No.	Time	Source	Destination	Protocol	Length Info	
41252	3334.6629726...	8c:fd:de:87:bf:41	HonHaiPr_cc:f7:eb	ARP	60	192.168.0.1 is at 8c:fd:de:87:bf:41
42401	3359.6171471...	8c:fd:de:87:bf:41	HonHaiPr_cc:f7:eb	ARP	60	Who has 192.168.0.57 Tell 192.168.0.1
42402	3359.6171583...	HonHaiPr_cc:f7:eb	8c:fd:de:87:bf:41	ARP	42	192.168.0.5 is at 38:b1:db:cc:f7:eb
43200	3402.4999007...	HonHaiPr_cc:f7:eb	8c:fd:de:87:bf:41	ARP	42	Who has 192.168.0.17 Tell 192.168.0.5
43201	3402.5067295...	8c:fd:de:87:bf:41	HonHaiPr_cc:f7:eb	ARP	60	192.168.0.1 is at 8c:fd:de:87:bf:41
43239	3405.6974959...	8c:fd:de:87:bf:41	HonHaiPr_cc:f7:eb	ARP	60	Who has 192.168.0.57 Tell 192.168.0.1
43240	3405.6975206...	HonHaiPr_cc:f7:eb	8c:fd:de:87:bf:41	ARP	42	192.168.0.5 is at 38:b1:db:cc:f7:eb
43379	3428.3519101...	HonHaiPr_cc:f7:eb	8c:fd:de:87:bf:41	ARP	42	Who has 192.168.0.17 Tell 192.168.0.5
43380	3428.3686363...	8c:fd:de:87:bf:41	HonHaiPr_cc:f7:eb	ARP	60	192.168.0.1 is at 8c:fd:de:87:bf:41
44488	3483.5218678...	8c:fd:de:87:bf:41	HonHaiPr_cc:f7:eb	ARP	60	Who has 192.168.0.57 Tell 192.168.0.1
44489	3483.5218917...	HonHaiPr_cc:f7:eb	8c:fd:de:87:bf:41	ARP	42	192.168.0.5 is at 38:b1:db:cc:f7:eb
49501	3528.5821824...	8c:fd:de:87:bf:41	HonHaiPr_cc:f7:eb	ARP	60	Who has 192.168.0.57 Tell 192.168.0.1
49502	3528.5821997...	HonHaiPr_cc:f7:eb	8c:fd:de:87:bf:41	ARP	42	192.168.0.5 is at 38:b1:db:cc:f7:eb
51677	3564.4184458...	8c:fd:de:87:bf:41	HonHaiPr_cc:f7:eb	ARP	60	Who has 192.168.0.57 Tell 192.168.0.1
51678	3564.4184624...	HonHaiPr_cc:f7:eb	8c:fd:de:87:bf:41	ARP	42	192.168.0.5 is at 38:b1:db:cc:f7:eb
54901	3605.8906096...	8c:fd:de:87:bf:41	HonHaiPr_cc:f7:eb	ARP	60	Who has 192.168.0.57 Tell 192.168.0.1
54902	3605.8906742...	HonHaiPr_cc:f7:eb	8c:fd:de:87:bf:41	ARP	42	192.168.0.5 is at 38:b1:db:cc:f7:eb
60378	3647.3604465...	8c:fd:de:87:bf:41	HonHaiPr_cc:f7:eb	ARP	60	Who has 192.168.0.57 Tell 192.168.0.1
60379	3647.3604591...	HonHaiPr_cc:f7:eb	8c:fd:de:87:bf:41	ARP	42	192.168.0.5 is at 38:b1:db:cc:f7:eb
62852	3682.1762050...	8c:fd:de:87:bf:41	HonHaiPr_cc:f7:eb	ARP	60	Who has 192.168.0.57 Tell 192.168.0.1
62853	3682.1762186...	HonHaiPr_cc:f7:eb	8c:fd:de:87:bf:41	ARP	42	192.168.0.5 is at 38:b1:db:cc:f7:eb
65479	3724.6722608...	8c:fd:de:87:bf:41	HonHaiPr_cc:f7:eb	ARP	60	Who has 192.168.0.57 Tell 192.168.0.1
65480	3724.6722850...	HonHaiPr_cc:f7:eb	8c:fd:de:87:bf:41	ARP	42	192.168.0.5 is at 38:b1:db:cc:f7:eb
65642	3727.3599010...	HonHaiPr_cc:f7:eb	8c:fd:de:87:bf:41	ARP	42	Who has 192.168.0.17 Tell 192.168.0.5
65643	3727.3670179...	8c:fd:de:87:bf:41	HonHaiPr_cc:f7:eb	ARP	60	192.168.0.1 is at 8c:fd:de:87:bf:41
65804	3754.8816700...	8c:fd:de:87:bf:41	HonHaiPr_cc:f7:eb	ARP	60	Who has 192.168.0.57 Tell 192.168.0.1
65805	3754.8816964...	HonHaiPr_cc:f7:eb	8c:fd:de:87:bf:41	ARP	42	192.168.0.5 is at 38:b1:db:cc:f7:eb
65947	3800.4480543...	8c:fd:de:87:bf:41	HonHaiPr_cc:f7:eb	ARP	60	Who has 192.168.0.57 Tell 192.168.0.1
65948	3800.4480792...	HonHaiPr_cc:f7:eb	8c:fd:de:87:bf:41	ARP	42	192.168.0.5 is at 38:b1:db:cc:f7:eb
66038	3828.6079053...	8c:fd:de:87:bf:41	HonHaiPr_cc:f7:eb	ARP	60	Who has 192.168.0.57 Tell 192.168.0.1
66039	3828.6079210...	HonHaiPr_cc:f7:eb	8c:fd:de:87:bf:41	ARP	42	192.168.0.5 is at 38:b1:db:cc:f7:eb
66250	3860.7350687...	HonHaiPr_cc:f7:eb	8c:fd:de:87:bf:41	ARP	42	Who has 192.168.0.17 Tell 192.168.0.5
66251	3860.7500384...	8c:fd:de:87:bf:41	HonHaiPr_cc:f7:eb	ARP	60	192.168.0.1 is at 8c:fd:de:87:bf:41
67595	3919.3599243...	HonHaiPr_cc:f7:eb	8c:fd:de:87:bf:41	ARP	42	Who has 192.168.0.17 Tell 192.168.0.5
67596	3919.3671534...	8c:fd:de:87:bf:41	HonHaiPr_cc:f7:eb	ARP	60	192.168.0.1 is at 8c:fd:de:87:bf:41
67730	3947.4000673...	8c:fd:de:87:bf:41	HonHaiPr_cc:f7:eb	ARP	60	Who has 192.168.0.57 Tell 192.168.0.1
67731	3947.4000902...	HonHaiPr_cc:f7:eb	8c:fd:de:87:bf:41	ARP	42	192.168.0.5 is at 38:b1:db:cc:f7:eb

Resp: FIG

Exercício (3): ARP

- Qual é a diferença de usarmos os filtros “**arp.opcode == 1**” e “**arp.opcode == 2**”?

Exercício Resolvido (8): ARP

- O que acontece se usarmos o filtro “arp.dst.proto_ipv4 == seu-ip”?

Exercício Resolvido (8): ARP

- O que acontece se usarmos o filtro “arp.dst.proto_ipv4 == seu-ip”?

No.	Time	Source	Destination	Protocol	Length	Info
77247	4530.9400302...	XiaomiCo_e2:bf:5f	Broadcast	ARP	60	Who has 192.168.0.234? Tell 192.168.0.2
77248	4530.9416241...	XiaomiCo_e2:bf:5f	Broadcast	ARP	60	Who has 192.168.0.235? Tell 192.168.0.2
77249	4530.9426609...	XiaomiCo_e2:bf:5f	Broadcast	ARP	60	Who has 192.168.0.236? Tell 192.168.0.2
77250	4530.9437502...	XiaomiCo_e2:bf:5f	Broadcast	ARP	60	Who has 192.168.0.237? Tell 192.168.0.2
77252	4530.9448282...	XiaomiCo_e2:bf:5f	Broadcast	ARP	60	Who has 192.168.0.238? Tell 192.168.0.2
77253	4530.9466327...	XiaomiCo_e2:bf:5f	Broadcast	ARP	60	Who has 192.168.0.239? Tell 192.168.0.2
77254	4530.9530757...	XiaomiCo_e2:bf:5f	Broadcast	ARP	60	Who has 192.168.0.240? Tell 192.168.0.2
77255	4530.9541392...	XiaomiCo_e2:bf:5f	Broadcast	ARP	60	Who has 192.168.0.241? Tell 192.168.0.2
77256	4530.9551841...	XiaomiCo_e2:bf:5f	Broadcast	ARP	60	Who has 192.168.0.242? Tell 192.168.0.2
77257	4530.9562700...	XiaomiCo_e2:bf:5f	Broadcast	ARP	60	Who has 192.168.0.243? Tell 192.168.0.2
77258	4530.9573250...	XiaomiCo_e2:bf:5f	Broadcast	ARP	60	Who has 192.168.0.244? Tell 192.168.0.2
77259	4530.9583813...	XiaomiCo_e2:bf:5f	Broadcast	ARP	60	Who has 192.168.0.245? Tell 192.168.0.2
77260	4530.9594160...	XiaomiCo_e2:bf:5f	Broadcast	ARP	60	Who has 192.168.0.246? Tell 192.168.0.2
77261	4530.9605095...	XiaomiCo_e2:bf:5f	Broadcast	ARP	60	Who has 192.168.0.247? Tell 192.168.0.2
77262	4530.9615083...	XiaomiCo_e2:bf:5f	Broadcast	ARP	60	Who has 192.168.0.248? Tell 192.168.0.2
77263	4530.9630192...	XiaomiCo_e2:bf:5f	Broadcast	ARP	60	Who has 192.168.0.249? Tell 192.168.0.2
77264	4530.9640142...	XiaomiCo_e2:bf:5f	Broadcast	ARP	60	Who has 192.168.0.250? Tell 192.168.0.2
77265	4530.9651880...	XiaomiCo_e2:bf:5f	Broadcast	ARP	60	Who has 192.168.0.251? Tell 192.168.0.2
77266	4530.9662306...	XiaomiCo_e2:bf:5f	Broadcast	ARP	60	Who has 192.168.0.252? Tell 192.168.0.2
77267	4530.9728934...	XiaomiCo_e2:bf:5f	Broadcast	ARP	60	Who has 192.168.0.253? Tell 192.168.0.2
77268	4530.9739173...	XiaomiCo_e2:bf:5f	Broadcast	ARP	60	Who has 192.168.0.254? Tell 192.168.0.2
78401	4547.4543980...	8c:fd:de:87:bf:41	HonHaiPr_cc:f7:eb	ARP	60	Who has 192.168.0.5? Tell 192.168.0.1
78402	4547.4544245...	HonHaiPr_cc:f7:eb	8c:fd:de:87:bf:41	ARP	42	192.168.0.5 is at 38:b1:db:cc:f7:eb
78507	4589.4409386...	8c:fd:de:87:bf:41	HonHaiPr_cc:f7:eb	ARP	60	Who has 192.168.0.5? Tell 192.168.0.1
78508	4589.4409481...	HonHaiPr_cc:f7:eb	8c:fd:de:87:bf:41	ARP	42	192.168.0.5 is at 38:b1:db:cc:f7:eb
78680	4668.2864757...	8c:fd:de:87:bf:41	HonHaiPr_cc:f7:eb	ARP	60	Who has 192.168.0.5? Tell 192.168.0.1
78681	4668.2864916...	HonHaiPr_cc:f7:eb	8c:fd:de:87:bf:41	ARP	42	192.168.0.5 is at 38:b1:db:cc:f7:eb
78729	4681.1913490...	XiaomiCo_e2:bf:5f	HonHaiPr_cc:f7:eb	ARP	60	Who has 192.168.0.5? Tell 192.168.0.2
78730	4681.1913738...	HonHaiPr_cc:f7:eb	XiaomiCo_e2:bf:5f	ARP	42	192.168.0.5 is at 38:b1:db:cc:f7:eb
78808	4710.2716217...	8c:fd:de:87:bf:41	HonHaiPr_cc:f7:eb	ARP	60	Who has 192.168.0.5? Tell 192.168.0.1
78809	4710.2716660...	HonHaiPr_cc:f7:eb	8c:fd:de:87:bf:41	ARP	42	192.168.0.5 is at 38:b1:db:cc:f7:eb
79199	4775.2970592...	8c:fd:de:87:bf:41	HonHaiPr_cc:f7:eb	ARP	60	Who has 192.168.0.5? Tell 192.168.0.1
79200	4775.2970725...	HonHaiPr_cc:f7:eb	8c:fd:de:87:bf:41	ARP	42	192.168.0.5 is at 38:b1:db:cc:f7:eb
80658	4831.6135078...	8c:fd:de:87:bf:41	HonHaiPr_cc:f7:eb	ARP	60	Who has 192.168.0.5? Tell 192.168.0.1
80659	4831.6135242...	HonHaiPr_cc:f7:eb	8c:fd:de:87:bf:41	ARP	42	192.168.0.5 is at 38:b1:db:cc:f7:eb
80928	4883.8374557...	8c:fd:de:87:bf:41	HonHaiPr_cc:f7:eb	ARP	60	Who has 192.168.0.5? Tell 192.168.0.1
80929	4883.8374796...	HonHaiPr_cc:f7:eb	8c:fd:de:87:bf:41	ARP	42	192.168.0.5 is at 38:b1:db:cc:f7:eb

Exercício (4): DHCP

- O que acontece se usarmos o filtro “dhcp”?

ICMPv6

- Mesmas funcionalidades do ICMPv4
 - Reportar erros no processamento de pacotes
 - Realizar diagnósticos
 - Enviar mensagens sobre as características da rede
- Assume as funções dos protocolos IPv4:
 - ARP: mapear um endereço físico a partir do lógico (IP)
 - RARP: mapear um endereço lógico a partir do físico
 - IGMP: gerenciamento de membros de grupos multicast

Exercício (5): ICMPv6

- O que acontece se usarmos o filtro “icmpv6”?